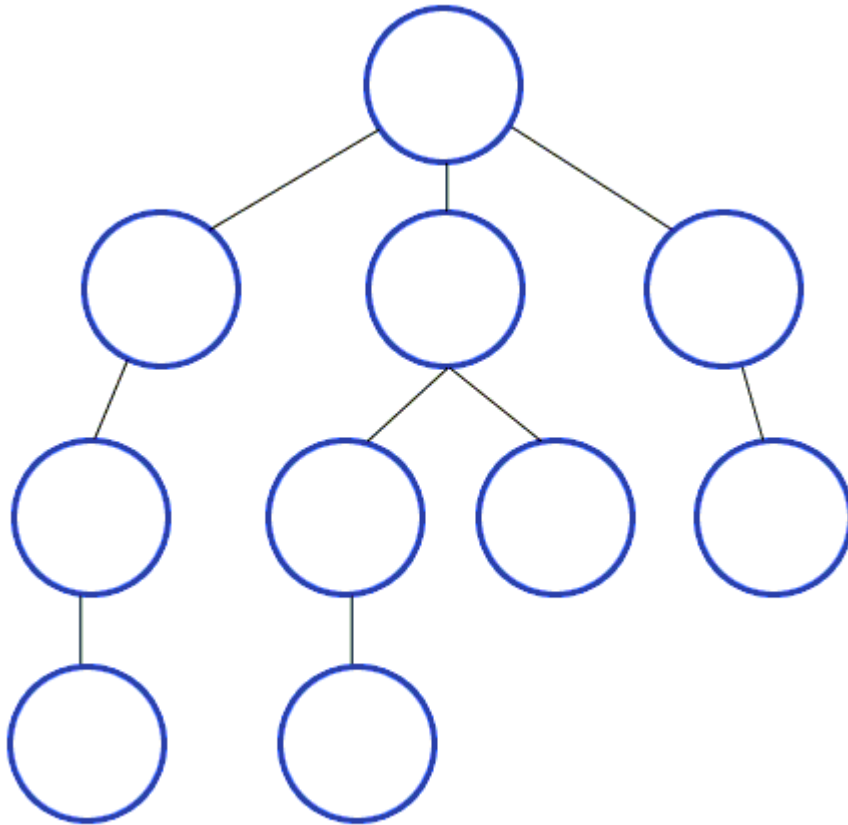
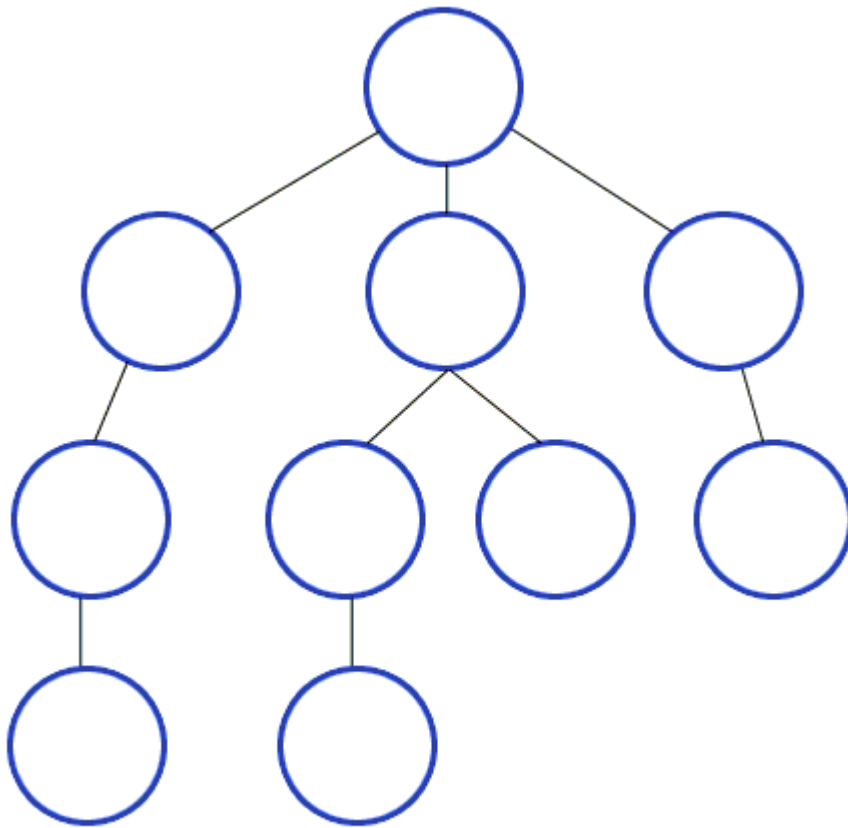


Depth-first search (DFS) is an algorithm for traversing or searching tree or graph data structures. One starts at the root (selecting some arbitrary node as the root in the case of a graph) and explores as far as possible along each branch before backtracking.



Breadth-first search (BFS) is an algorithm for traversing or searching tree or graph data structures. It starts at the tree root (or some arbitrary node of a graph, sometimes referred to as a 'search key') and explores the neighbor nodes first, before moving to the next level neighbors.



Pseudocode

```
BFS(root)
  Pre: root is the node of the BST
  Post: the nodes in the BST have been visited in breadth first order
  q ← queue
  while root = ∅
    yield root.value
    if root.left = ∅
      q.enqueue(root.left)
    end if
    if root.right = ∅
      q.enqueue(root.right)
    end if
    if !q.isEmpty()
      root ← q.dequeue()
    else
      root ← ∅
    end if
  end while
end BFS
```